

Astrophysics: PH 556

Lecture Notes

Soham Sahasrabuddhe
Department of Physics
Indian Institute of Technology, Bombay

Fall 2025

Contents

1	Introduction	2
2	Magnitudes, Fluxes and Colours	2
2.1	Flux	2
2.2	Magnitude	2
2.3	Filters	3
2.4	Absolute Magnitude	3
3	Sources of Star Energy	4
3.1	Triple Alpha Process	4
4	Star Formation	5
5	Radio Astronomy	5
5.1	History	5
5.2	Radio Sources	5
5.3	Radio Surveys	7
5.4	Fluxes	7
6	Radio Telescopes	7
7	Orbits	8
8	Coordinate Systems	9
8.1	Horizontal Coordinate System	9
8.2	Equatorial System	10
8.3	HA-Dec Coordinate System	10
8.4	Spherical Coordinate System	10
8.5	Sine Rule	10
8.6	Cosine Rule	11

1 Introduction

This document contains the lecture notes for the course **Astrophysics: PH 556**, taught by **Prof. Varun Bhalerao** at the Indian Institute of Technology, Bombay, in Fall 2025. This course covers various topics in astrophysics, including stellar dynamics, galactic structure, cosmology, and high-energy astrophysics, with main focus on observational astronomy + data analysis.

2 Magnitudes, Fluxes and Colours

Typical human eye response is in the range of 400 nm to 700 nm. The response is not uniform across this range, and the eye is most sensitive to light at around 555 nm, which corresponds to the green part of the spectrum. The human eye's resolution is about 1 arcmin, which means it can distinguish two points in the sky that are separated by at least this angle. On the contrary, the size of the nearest star, assuming it is of one solar mass ($M_\odot$), is about $0''.005$, which is much smaller than the eye's resolution limit. This means that the eye cannot resolve individual stars in a cluster or galaxy, and instead sees them as a single point of light.

Now what properties of a star can we observe with the human eye?

- Colours
- Brightness: More energy received $\implies$ more photons received, in units of Luminosity (L), which means energy emitted per unit time. *Brightness is relativistically invariant over distances.*
- Position @ time of observation
- Size
- Twinkling

2.1 Flux

Definition 1. Power received per unit area, for a uniform spherical source at distance d is given by-

$$\phi = \frac{P}{4\pi d^2}$$

Whereas power received per unit area per unit frequency is flux density. Also called specific flux.

Specific intensity (or surface brightness) is the power received (or emitted) per unit area per unit frequency per unit solid angle.

$$F_\nu = \int I_\nu \cos \theta d\Omega$$

It is independent of distance.

2.2 Magnitude

Definition 2. The magnitude of a star is a measure of its brightness as seen from Earth. It is defined such that a difference of 5 magnitudes corresponds to a factor of 100 in brightness.

$$m = -2.5 \log \left(\frac{F}{F_0} \right) + C$$

where F is the flux received from the star, F_0 is a reference flux, and C is a constant that can be adjusted based on the system of magnitudes used. This implies brighter stars have lower magnitudes. The magnitude scale is logarithmic, so a difference of 1 magnitude corresponds to a brightness change of about 2.512 times (the fifth root of 100).

Remark 1. *For comparison, one can derive a relation:*

$$m_1 - m_2 = -2.5 \log \left(\frac{F_1}{F_2} \right)$$

This means that if you know the magnitudes of two stars, you can find the ratio of their fluxes. If $m_1 < m_2$, then $F_1 > F_2$, meaning the first star is brighter than the second.

For reference, the Sun has an apparent magnitude of about -27, Sirius (the brightest star in the night sky) has an apparent magnitude of about -1.46, full moon has an apparent magnitude of about -12.74, and the faintest stars visible to the naked eye under ideal conditions have an apparent magnitude of about +6.5. The Growth India Telescope can observe objects with magnitudes up to about +21.

2.3 Filters

Commonly used filters in astronomy are: UBVRI, by Johnson. Detects near UV, visible, and near-infrared light. Each filter allows a specific range of wavelengths to pass through, which helps in studying the properties of celestial objects. SDSS is another commonly used system, which includes filters u, g, r, i, z. These filters are designed to cover the ultraviolet to near-infrared spectrum and are used in the Sloan Digital Sky Survey.

2.4 Absolute Magnitude

Definition 3. *The absolute magnitude of a star is the apparent magnitude it would have if it were placed at a distance of 10 parsecs (about 32.6 light-years) from Earth. It is a measure of the intrinsic brightness of a star.*

$$M = m - 5 \log \left(\frac{d}{10} \right)$$

where M is the absolute magnitude, m is the apparent magnitude, and d is the distance to the star in parsecs.

Planck's law for blackbody radiation gives the spectral energy distribution against wavelength, given by:

$$B(\lambda, T) = \frac{2hc^2}{\lambda^5} \frac{1}{e^{hc/\lambda kT} - 1}$$

where $B(\lambda, T)$ is the intensity of radiation at wavelength λ and temperature T , h is Planck's constant, c is the speed of light, and k is Boltzmann's constant. This law describes how the intensity of radiation emitted by a blackbody varies with wavelength and temperature.

The peak of the distribution shifts to higher frequencies as the temperature increases, which is described by Wien's displacement law:

$$\lambda_{\max} = \frac{b}{T}$$

where $\lambda_{\max}$ is the wavelength at which the emission is maximized, T is the temperature of the blackbody, and b is Wien's displacement constant (approximately 2.897×10^{-3} m K).

Commonly used tool for studying astro physics is the Hertzsprung-Russell (HR) diagram, which plots stars according to their absolute magnitude (or luminosity) against their effective temperature (or spectral class). The temperature is figured out from the difference in magnitudes of B and R filters (B-R).

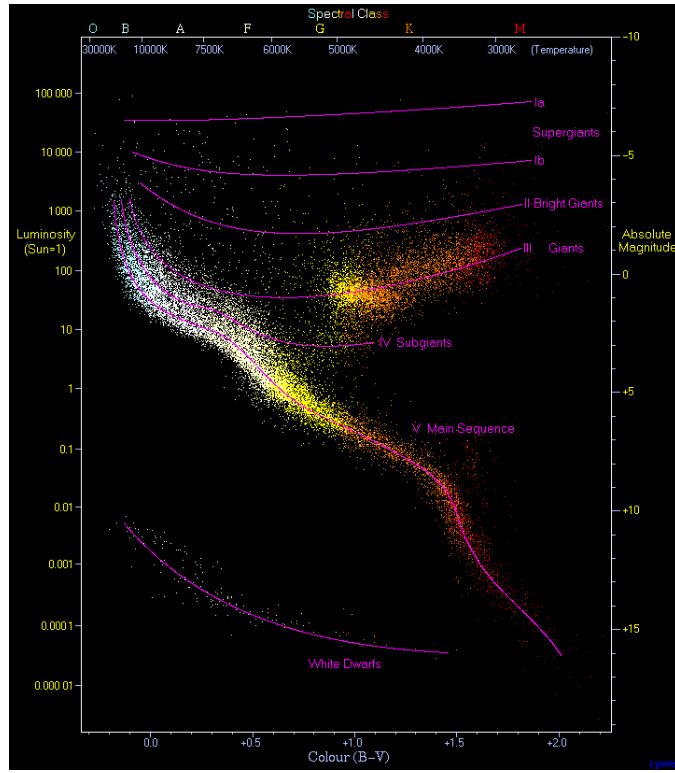
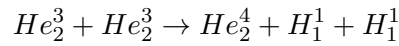
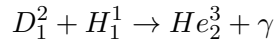
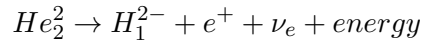
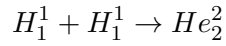


Figure 1: HR diagram

3 Sources of Star Energy

Stars generate energy through nuclear fusion, primarily converting hydrogen into helium in their cores. This process releases a tremendous amount of energy, which is radiated away as light and heat.



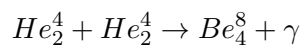
The energy produced in the core of a star is transported to the surface through radiation and convection. This is the PP chain reaction, which is the dominant energy source in stars like the Sun. This process is called collisional ionization, where the high temperatures and pressures in the star's core allow protons to overcome their electrostatic repulsion and fuse together.

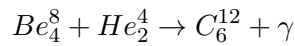
$$\frac{3}{2}kT = 13.6\text{ eV}$$

This is the energy required to ionize hydrogen, using this we calculate the temperature of the core to be $T = 10^5 K$. Rate is $\propto \rho T^4$, where ρ is the density of the star.

3.1 Triple Alpha Process

In more massive stars, once hydrogen is depleted, helium can fuse into heavier elements through the triple-alpha process:





This leads to formation of carbon. The process is called the triple-alpha process because it involves three helium nuclei (alpha particles) fusing together to form carbon. This process occurs in the cores of medium stars, where temperatures and pressures are high enough to allow for the fusion of helium into heavier elements. Rate of this process is $\propto \rho^2 T^{40}$. This process is now followed by fusion of carbon into heavier elements, such as oxygen and nitrogen, known as CNO cycle. The fusion of these heavier elements continues until iron is formed, which is the end point of fusion in stars because fusing iron does not release energy. Rate is $\propto \rho T^{17}$.

4 Star Formation

Imagine a spherical cloud of $100 M_\odot$, under gravitational collapse. There can be multiple possibilities for such a system, for instance, some local disturbance can cause formation of multiple stars, or a single star can form. The Jean's mass is the critical mass above which a cloud will collapse under its own gravity. This is called the Jean's Criteria. Mass can be lost as well, for instance, through radiation pressure or stellar winds. Jean's mass is criteria due to inhomogeneity in the cloud, if temperature is high, or there is turbulence, all this factors can affect the stability of the cloud, and hence the Jean's Mass magnitude.

Definition 4. *The Jean's mass is the mass that a spherical cloud of interstellar gas must have in order to contract under its own weight.* https://en.wikipedia.org/wiki/Jean's_instability

Cool video displayed in the class: <http://www.astro.ex.ac.uk/people/mbate/Cluster/> Check out slides on moodle for more!

5 Radio Astronomy

5.1 History

Optical one is oldest, radio started with an accident. Karl Jansky, a radio engineer, was investigating static in radio communications when he discovered that the static was coming from the Milky Way. He built a large antenna to study this radiation at 20.5 MHz, which led to the first detection of cosmic radio waves in 1932. This discovery laid the foundation for radio astronomy. Grote Reber, an American engineer, built the first radio telescope in his backyard in 1937, which was a parabolic dish that could detect radio waves from space. He published his findings in 1940, marking the beginning of systematic radio astronomy. World War II saw significant advancements in radar technology, which were later adapted for radio astronomy. After the war, radio telescopes became larger and more sophisticated, leading to the discovery of pulsars, quasars, and cosmic microwave background radiation. Indian Radio Astronomy began with Govind Swarup, who started the first radio telescope in India, with help of Tata Institute of Fundamental Research (TIFR) in 1960. The Ooty Radio Telescope, built in the 1970s, is one of the largest radio telescopes in Asia and has made significant contributions to the field.

5.2 Radio Sources

The optical sky is dominated by stars, ionised gas and dust (absorption). Radio sky is dominated by pulsars, supernova remnants, star forming regions, and active galactic nuclei (AGN) and galaxies (example NGC 5128, Sagittarius A*). Pulsars are rapidly rotating neutron stars that emit beams of radio waves, while supernova remnants are the remnants of exploded stars that emit radio waves as they expand. Star forming regions are areas where new stars are being born, and AGN are supermassive black holes at the centers of galaxies that emit large amounts of radio waves. The Crab Nebula is very bright in the radio sky. Some of the brightest radio sources in the sky include Cassiopeia A, Cygnus A, Orion A and Centaurus A.

Remark 2. *Bright Radio Sources are faint in the optical sky and vice versa. Why?*

This is because radio waves can penetrate dust and gas that block optical light, allowing us to see objects that are otherwise hidden in the optical spectrum. Additionally, they emit strong radio waves due to their high-energy processes.

Dominant physical mechanism for continuum radio emission is due to synchrotron emission by relativistic charged particles in magnetic fields. This occurs when charged particles, such as electrons, spiral around magnetic field lines, emitting radiation in the radio frequency range. The intensity of this emission depends on the strength of the magnetic field and the energy of the particles.

Lemma 1. *The characteristic frequency of synchrotron radiation is given by:*

$$\nu_c = \frac{3\gamma^2 e B}{4\pi m_e c} \approx 4.2 \times 10^6 \gamma^2 (B/1G) \text{ Hz}$$

where ν_c is the characteristic frequency, γ is the Lorentz factor of the electrons, e is the charge of the electron, B is the magnetic field strength, m_e is the mass of the electron.

If the electron energy distribution is a power law, the spectral energy distribution is a power law as well. The power is given by the relation:

$$P = \frac{4e^4 B^2 \gamma^2}{9m_e^2 c^3}$$

Thermal Bremsstrahlung is another mechanism for radio emission, which occurs when electrons interact with ions in a hot plasma, emitting radiation as they decelerate. This process is significant in hot ionized gas, such as that found in star-forming regions and supernova remnants.

Lemma 2. *The intensity of thermal bremsstrahlung emission depends on the temperature and density of the plasma. Free free emission coefficient is given by:*

$$j_\nu \propto n_e^2 T_e^{-\frac{1}{2}} \exp\left(-\frac{h\nu}{kT_e}\right)$$

where j_ν is the emission coefficient, n_e is the electron density, T_e is the electron temperature.

Synchrotron emission is linearly polarised perpendicular to the magnetic field, theoretical maximum: 70%. Polarised radio waves rotate as they pass through magnetized plasma:

$$\phi = RM \times \lambda^2$$

where Rotation Measure (RM) is a measure of the amount of Faraday rotation that has occurred.

- Pulsars: Highly magnetised, rotating neutron stars that emit beams of radiation. They are detected as periodic pulses of radio waves, with periods ranging from milliseconds to seconds. Have very strong magnetic fields, size reduced from millions of km to about 10 km, and are very dense, with masses around $1.4 M_\odot$.
- Around luminous stars from accelerated winds (charged particles) around star forming regions. Similar to solar wind, but much stronger. These winds can interact with the interstellar medium.
- Around supernovae, where the star explodes and ejects material at high speeds, creating a shock wave that accelerates particles and produces radio emission. The Crab Nebula is a well-known example of a supernova remnant that emits strong radio waves. The magnetic field is weak but ubiquitous in this case.
- From Supermassive black holes at centre of galaxies, quasars and blazars. From microquasars, which are smaller versions of quasars, and emit jets of particles at relativistic speeds. These jets can produce strong radio emission as they interact with the surrounding medium. The jets are highly collimated and can extend for thousands of light-years.

5.3 Radio Surveys

- NVSS: NRAO VLA Sky Survey, 1.4 GHz, 45% sky coverage, 0.45 mJy sensitivity, 1.8 million sources.
- FIRST: Faint Images of the Radio Sky at Twenty-Centimeters, 1.4 GHz, 10% sky coverage, 0.15 mJy sensitivity, high resolution
- LOFAR: Low Frequency Array (120 - 240 MHz), 100% sky coverage, 0.1 mJy sensitivity.
- VLASS: VLA Sky Survey, 3 GHz, 100% sky coverage, 0.1 mJy sensitivity, in construction.

Exoplanet Detection:

- Auroral Emission like Jupiter
- Cyclotron radiation from magnetosphere

5.4 Fluxes

1 Jansky (Jy) = $10^{-26} \text{ W m}^{-2} \text{ Hz}^{-1}$. A 1 Jansky is a quite bright source according to modern standards. A electron orbiting a proton with parallel spins has higher energy than if the spins are anti-parallel. The photon emitted when transitioning from higher to lower energy state has a frequency of 1.4 GHz, which is the frequency of the 21 cm line of hydrogen. This line is used in radio astronomy to study the distribution and motion of hydrogen in galaxies and the interstellar medium. This HI can be used to trace neutral hydrogen and to measure its kinematics. We can measure the rotation speed of object using the Doppler shift of the 21 cm line. The rotation curve of a galaxy can be obtained by measuring the Doppler shift of the 21 cm line at different positions along the galaxy's disk. This allows us to determine the rotation speed of the galaxy as a function of radius, which is crucial for understanding its mass distribution and dynamics.

6 Radio Telescopes

GMRT (Giant Metrewave Radio Telescope) is a radio telescope array located in India, consisting of 30 antennas spread over an area of 25 km. It operates at frequencies ranging from 150 MHz to 1.5 GHz. The GMRT is one of the largest and most sensitive radio telescopes in the world. GMRT has wire mesh as a reflector, utilising the large wavelength of radio waves. Arecibo Observatory was a large radio telescope located in Puerto Rico, which was operational from 1963 until its collapse in 2020. It had a dish diameter of 305 meters. The Very Large Array (VLA) is a radio telescope array located in New Mexico, USA, consisting of 27 antennas arranged in a Y-shaped configuration. Each antenna has a diameter of 25 meters and operates at frequencies ranging from 74 MHz to 50 GHz. The VLA is one of the most versatile and powerful radio telescopes in the world.

Angular resolution of a telescope: Three cases-

- Airy disk: Can separate two sources if they are separated by an angle greater than the diffraction limit, which is given by the Rayleigh criterion:

$$\mathcal{R} = 1.22 \frac{\lambda}{D}$$

where θ is the angular resolution, λ is the wavelength of the observed radiation, and D is the diameter of the telescope's aperture.

- Rayleigh Limit: Minimum separation between two point sources that can be resolved by a telescope.
- Not observable

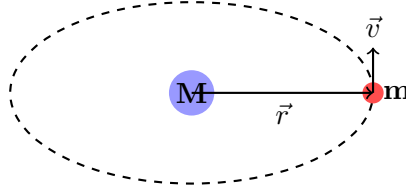
Theorem 1. *Van Cittert Zernike Theorem states that the visibility of a source is related to the Fourier transform of the intensity distribution of the source. This theorem is fundamental in radio interferometry, where multiple telescopes work together to achieve higher resolution by combining their signals. The visibility function is given by:*

$$V(u, v) = \langle E(r_1)E(r_2) \rangle$$

$$V(u, v) = \mathbb{F}(E(r_1)E(r_2)^*)$$

7 Orbits

Central body of mass M (massive), a smaller body of mass m rotating around it an arbitrary orbit with velocity $\vec{v}$,



$$\vec{a} = \frac{GM}{|\vec{r}|^2}$$

We need to consider the energy of the system for bounded/unbounded states, if:

- $E < 0$: System is bounded, no escape, orbit formation
- $E = 0$: Escape Velocity case,

$$v_{esc} = \sqrt{\frac{2GM}{r}}$$

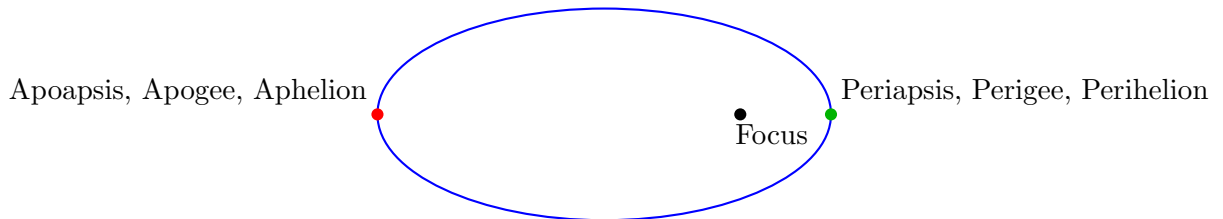
- $E > 0$: Velocity greater than escape velocity, we escape the system, results in unbounded orbits.

Note: While solving for orbits, we consider point masses and assume they don't collide during the motion.

Remark 3. *Mathematical Coincidence, Radius of a non-spinning blackhole (Schwarzschild) can be calculated without relativity,*

$$R_s = \frac{2GM}{c^2}$$

State Vector to lightspeed engine reference from Captain Marvel, you only need seven parameters to determine a orbit: $t_0, \vec{r}_0, \vec{v}_0$.



To visualise orbit configurations, cool website <https://orbitalmechanics.info/> The seven parameters for ellipse are: t_0 (epoch), a (semi major axis), e (eccentricity), ω (argument of periapsis), i (inclination), Ω (longitude of ascending node), and ν (true anomaly at epoch).

Definition 5. *Kepler's Third Law says*

$$T^2 \propto r^3$$

This gives for elliptical object-

$$v(r) = \sqrt{GM\left(\frac{2}{r} - \frac{1}{a}\right)}$$

with total energy given as in terms of semi-major axis

$$E_{total} = -\frac{GM}{2a}$$

For a binary system, the relation transforms into

$$T^2 = \frac{4\pi^2 a^3}{G(M_1 + M_2)}$$

Exercise

Calculate all elliptical parameters for Sun-Earth system for circular orbit, and elliptical orbit with $e = 0.0167$.

8 Coordinate Systems

Astronomical coordinate systems are used to specify the position of celestial objects in the sky. The most commonly used systems are the horizontal (alt-azimuth), equatorial, and ecliptic coordinate systems.

8.1 Horizontal Coordinate System

The horizontal coordinate system is a local, observer-based system that describes the position of objects relative to the observer's location on Earth. It uses two angles:

- **Azimuth (A):** The angle measured clockwise from the north point along the horizon to the point directly below the object. Azimuth ranges from 0° (north), through 90° (east), 180° (south), 270° (west), and back to 360° .
- **Altitude (h):** The angle above the horizon to the object, ranging from 0° (on the horizon) to $+90^\circ$ (at the zenith, directly overhead). Negative values indicate objects below the horizon (not visible).

The **horizon** is defined as $h = 0^\circ$, the **zenith** as $h = +90^\circ$, and the **nadir** as $h = -90^\circ$ (directly below the observer).

The horizontal system is time-dependent: as the Earth rotates, the azimuth and altitude of celestial objects change continuously.

Definition 6. *The culmination of a celestial object is the instant it attains its maximum altitude above the horizon during its diurnal motion. For objects that cross the local meridian this coincides with the object's transit across the meridian (due south for northern-hemisphere observers, due north for southern-hemisphere observers).*

For more exploration, visit <https://stellarium-web.org/>

8.2 Equatorial System

Defined by right ascension (RA) and declination (Dec), the equatorial coordinate system is fixed relative to the celestial sphere and is analogous to Earth's longitude and latitude. Declination is measured in degrees north (+) or south (-) of the celestial equator ($-90^\circ \leq \delta \leq +90^\circ$). Right ascension is measured eastward along the celestial equator in hours, minutes and seconds ($0^h \leq \alpha < 24^h$), where 24^h equals 360° .

Definition 7. North Celestial Pole:

The point where Earth's rotation axis, extended northward, intersects the celestial sphere; located at declination $+90^\circ$. The star Polaris lies close to this point at present.

Definition 8. South Celestial Pole:

The point where Earth's rotation axis, extended southward, meets the celestial sphere; located at declination -90° . There is no comparably bright star near this point.

Definition 9. Circumpolar stars:

Stars whose declination satisfies $|\delta| \geq 90^\circ - \phi$ for an observer at latitude ϕ . Such stars never set below the horizon and appear to circle the corresponding celestial pole.

8.3 HA-Dec Coordinate System

Definition 10. Local Sidereal Time (LST)

The local sidereal time at a given location is the right ascension of celestial objects that are currently crossing the observer's local meridian. It is the HA of Vernal Equinox as well. Need to use mod(24) logic, for LST's exceeding a day

The Hour Angle (H or HA) and Declination (Dec, δ) form a local celestial coordinate system tied to the observer's meridian. Declination is the angular distance north (+) or south (-) of the celestial equator (analogous to terrestrial latitude). The hour angle of an object is the angular distance, measured along the celestial equator, between the local meridian and the object's hour circle; it is related to right ascension (α) and local sidereal time (LST) by

$$H = \text{LST} - \alpha,$$

typically expressed in hours ($1h = 15^\circ$). By convention HA increases toward the west. The declination doesn't change with time since the distance of star from equatorial plane is same, at each position of orbit.

Definition 11. *An **equinox** in astronomical coordinate systems is a reference point on the celestial sphere, defined by the intersection of the celestial equator and the ecliptic. Standard equinoxes (e.g., B1950, J2000) specify the orientation of the coordinate system at a particular epoch and are used to account for precession and other motions of Earth's axis. The **epoch** is the specific date and time at which celestial coordinates or orbital elements are calculated.*

For more reference on Coordinate System, refer <https://bit.ly/426Vqpx>

8.4 Spherical Coordinate System

Define a reference plane passing through the center, which is a great circle. Consider three points on the surface of sphere, namely A, B, and Z. Consider great circles passing from AZ, ZB, and AB. The AZB region forms a spherical triangle. Now assume star is at A, North Celestial Pole is B and Z is the Zenith. Then

$$\angle ZBA = H, \angle BZA = 360 - A, a = 90 - h, z = 90 - \delta$$

8.5 Sine Rule

For a spherical triangle with sides a, b, c opposite to angles A, B, C respectively, the sine rule states that:

$$\frac{\sin A}{\sin a} = \frac{\sin B}{\sin b} = \frac{\sin C}{\sin c}$$

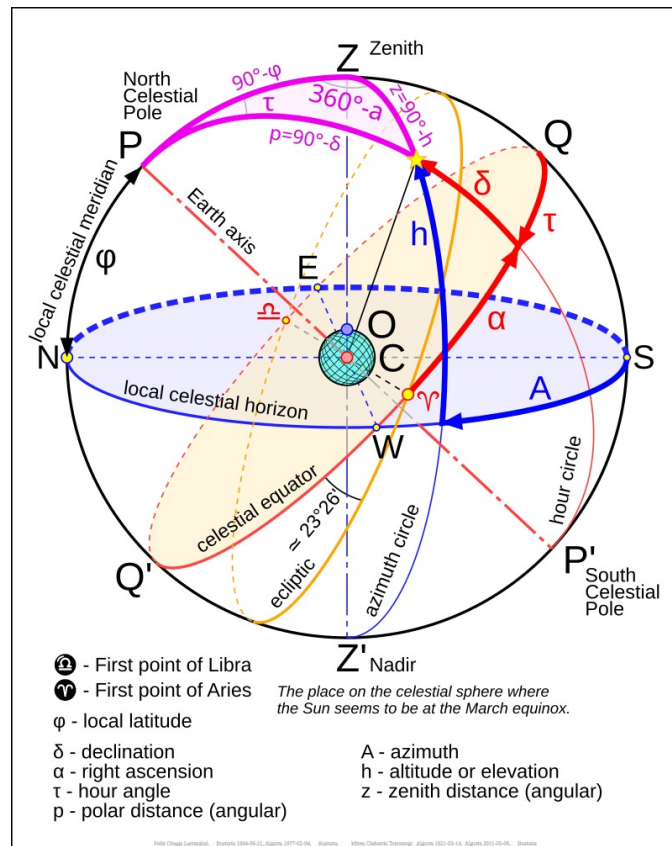


Figure 2: Celestial Sphere with Spherical Triangle

8.6 Cosine Rule

For a spherical triangle with sides a , b , c opposite to angles A , B , C respectively, the cosine rule states that:

$$\cos a = \cos b \cos c + \sin b \sin c \cos A$$

$$\cos A = -\cos B \cos C + \sin B \sin C \cos a$$

For more check,

- https://en.wikipedia.org/wiki/Spherical_trigonometry
- <https://science.nasa.gov/learn/basics-of-space-flight/chapter2-2/>
- https://astrogreg.com/convert_ra_dec_to_alt_az.html